ANSWER KEY & MARKING SCHEME SET 1

Chapter 4: **Carbon and its Compounds** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	Weak forces need little energy.
2	**(a)** $CH_4 + 2\,O_2 \longrightarrow CO_2 + 2\,H_2O$	Atoms of C, H and O balance.
3	**(c)** ethene	C_2H_4 = 28.
4	**(a)** Both A and R are true and R is the correct explanation of A.	Incomplete combustion gives soot.
5	**(b)** 6	Given in the equation.
6	**(a)** ethanoic acid	Vinegar.
7	**(c)** A is true but R is false.	A true; in graphite each carbon bonds to three others, so R is false.
8	**(d)** a saturated hydrocarbon	Saturated hydrocarbons burn cleanly.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Self-linking into long chains, branches and rings (catenation) together with four bonds per atom gives a huge number of compounds.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but R is an additional fact about another oxidising agent and does not explain A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$2\,C_2H_6 + 7\,O_2 \longrightarrow 4\,CO_2 + 6\,H_2O$ (1); Products: carbon dioxide and water (1).	1+1
12	$3\,n + 1 = 22$, n = 7 (1); C_7H_{16} (1).	1+1
13	Propyne has a triple bond between two carbon atoms (1); Two hydrogen molecules add, giving propane (1).	1+1
14	Molecular mass = 30 u (1); 24/30 x 100 = 80% (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$CH_3{-}CH{=}CH_2$ (1); Propene is unsaturated (a double bond); propane is saturated (1); Propene undergoes addition reactions; propane undergoes substitution (1).	1+1+1
16	n-butane: 3 C-C, 10 C-H (1); isobutane: 3 C-C, 10 C-H (1); Same number of bonds; isomers (1).	1+1+1
17	C1-C4 gases (1); C5-C17 liquids (1); Higher are solids (wax) (1).	1+1+1
	OR: $CH_4 + 2\,O_2 \longrightarrow CO_2 + 2\,H_2O$ (1); Clean blue flame (1); Heat and light are given out (exothermic) (1).	1+1+1
18	Graphite: each C bonded to 3, free electron (1); Diamond: each C bonded to 4, no free electrons (1); Fullerene: discrete C_{60} molecules (1).	1+1+1
	OR: Pentane C_5H_{12}: 72 u (1); Difference CH_2 = 14 u (1); Methane 16, ethane 30, propane 44, butane 58 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) C_5H_{12} (pentane) (1). (ii) Ethyne, C_2H_2 (1). (iii) 14 u (a CH_2 unit) (1). (iv) Propane C_3H_8 has $3n + 1 = 10$ covalent bonds (1); all carbon-carbon bonds are single bonds, so each carbon holds the maximum number of hydrogen atoms (saturated) (1).	1+1+1+2
20	$C_2H_5OH + 3\,O_2 \longrightarrow 2\,CO_2 + 3\,H_2O$ (1); Burns with a clean flame, leaving no residue (1); It is renewable, obtained from sugarcane (1); It dissolves in petrol (1); Reduces the use of petrol (1).	1+1+1+1+1
	OR: Diamond: 3D, hard (1); Graphite: layered, conducts (1); Fullerene: C_{60} (1); All burn to CO_2 (1); Same element (1).	1+1+1+1+1

Mistakes to avoid:

- Counting covalent bonds wrongly: an alkane C_nH_{2n+2} has $3n + 1$ bonds (include all C-H and C-C bonds).
- Confusing ethanol and ethanoic acid: ethanol (C_2H_5OH) is an alcohol, ethanoic acid (CH_3COOH) releases CO_2 with carbonates; ethanol does not.

- Naming the chain from the wrong carbon count, e.g. calling CH_3CH_2CHO 'propanol' instead of propanal; the suffix comes from the functional group (-ol, -al, -one, -oic acid).
- Writing an addition reaction for a saturated hydrocarbon; saturated compounds undergo substitution, unsaturated ones undergo addition.
- Saying that detergents form micelles but soaps do not, or that soap works well in hard water; soaps give scum in hard water.
- Stating that diamond and graphite conduct electricity; only graphite does (it has free electrons).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 4: **Carbon and its Compounds** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** carbon dioxide and water	Complete combustion of a carbon compound gives CO2 and H2O.
2	**(c)** methyl ethanoate	Ester of ethanoic acid.
3	**(a)** Both A and R are true and R is the correct explanation of A.	Partial ionisation.
4	**(c)** 75%	12/16 x 100 = 75%.
5	**(a)** unsaturated	High carbon percentage.
6	**(a)** Both A and R are true and R is the correct explanation of A.	Both explain the variety.
7	**(b)** 85.7%	24/28 = 85.7%.
8	**(d)** hydrocarbon tails	Tails cluster with grease.
9	**(c)** A is true but R is false.	Hydrogen adds across the double bonds, converting them into single bonds; so R is false.
10	**(d)** A is false but R is true.	No free electrons are available in diamond, so it does not conduct; A is false, R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Ethyl ethanoate (1); Fruity smell (1).	1+1
12	Methane has 75% C, ethyne 92% (1); Limited oxygen leaves unburnt carbon (1).	1+1
13	Carbon is not fully oxidised (1); Ethyne (highest carbon %) (1).	1+1
14	Lower energy per litre than petrol (1); Advantage: burns cleanly, renewable (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$CH_2{=}CH_2 + H_2 \longrightarrow CH_3CH_3$, addition (1); $CH_3CH_3 + Cl_2 \longrightarrow CH_3CH_2Cl + HCl$ (1); Substitution (sunlight) (1).	1+1+1
16	Ethyl ethanoate (1); Alcohol: ethanol (1); $CH_3COOC_2H_5 + H_2O \longrightarrow CH_3COOH + C_2H_5OH$ (1).	1+1+1
17	Hydration of ethene (1); $CH_2{=}CH_2 + H_2O \longrightarrow CH_3CH_2OH$ (1); Acid catalyst (1).	1+1+1
	OR: X: ethanoic acid (1); Product: ethyl ethanoate (1); $CH_3COOH + C_2H_5OH \longrightarrow CH_3COOC_2H_5 + H_2O$ (1).	1+1+1
18	Bromine water: ethene decolourises (1); Sodium bicarbonate: acid effervesces (1); Ethane and ethanol do not (1).	1+1+1
	OR: Do not form scum (1); Their salts with Ca/Mg are soluble (1); Work in acidic water too (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Carbon dioxide (1). (ii) It turns lime water milky (1). (iii) Ethyl ethanoate (an ester) (1). (iv) $CH_3COOH + C_2H_5OH \xrightarrow{\text{conc.}\ H_2SO_4} CH_3COOC_2H_5 + H_2O$ (1); esterification (1).	1+1+1+2
20	Soap: $C_{17}H_{35}COONa$ (1); Detergent: sulphonate salt (1); Micelle: tails inward (1); Hard water gives scum with soap (1); Detergent works in hard water (1).	1+1+1+1+1
	OR: $2\,CH_3COOH + Na_2CO_3 \longrightarrow 2\,CH_3COONa + H_2O + CO_2$ (1); Brisk effervescence is seen (1); The gas is carbon dioxide (1); It turns lime water milky (1); Sodium ethanoate is formed (1).	1+1+1+1+1

Mistakes to avoid:

- Counting covalent bonds wrongly: an alkane C_nH_{2n+2} has $3n + 1$ bonds (include all C-H and C-C bonds).
- Confusing ethanol and ethanoic acid: ethanol (C_2H_5OH) is an alcohol, ethanoic acid (CH_3COOH) releases CO_2 with carbonates; ethanol does not.
- Naming the chain from the wrong carbon count, e.g. calling CH_3CH_2CHO 'propanol' instead of propanal; the suffix comes from the functional group (-ol, -al, -one, -oic acid).
- Writing an addition reaction for a saturated hydrocarbon; saturated compounds undergo substitution, unsaturated ones undergo addition.

- Saying that detergents form micelles but soaps do not, or that soap works well in hard water; soaps give scum in hard water.
- Stating that diamond and graphite conduct electricity; only graphite does (it has free electrons).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 4: **Carbon and its Compounds** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** ethane	No acidic hydrogen.
2	**(c)** A is true but R is false.	A true; R false (it is saturated).
3	**(c)** 2	C2H2 to C2H4 to C2H6.
4	**(c)** ethyne	Triple bond is shortest.
5	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true, but R does not explain the smell.
6	**(d)** 3	n-, iso-, neopentane.
7	**(a)** ethanol is oxidised and $KMnO_4$ is reduced	$KMnO_4$ acts as oxidising agent.
8	**(a)** Both A and R are true and R is the correct explanation of A.	Functional group governs reactions.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Little energy is needed to overcome weak intermolecular forces.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true; R describes how esters are formed, not why they smell sweet.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Hydrogen (1); Burns with a pop sound (1).	1+1
12	Ethanol forms hydrogen bonds with water (1); Hexane is non-polar (1).	1+1
13	Soap is made from fats or vegetable oil with alkali (1); Mineral oil is hydrocarbon with no ester to hydrolyse (1).	1+1
14	Ethanol (1); Hydrogen bonding between -OH groups (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Saponification needs a base (1); Hydroxide attacks the ester (1); Gives soap (salt) which cannot reverse (1).	1+1+1
16	Removing four electrons needs huge energy (1); Gaining four electrons is also unfavourable (1); Carbon shares electrons (1).	1+1+1
17	Fat + NaOH gives soap + glycerol (1); Soap is a sodium salt of fatty acid (1); Salt reduces solubility so soap separates (1).	1+1+1
	OR: Distilled water: lather (1); Hard water: scum, little lather (1); Ca/Mg ions form insoluble salts (1).	1+1+1
18	Functional group decides reactions (1); -OH: alcohol reactions (1); -COOH: acidic character (1).	1+1+1
	OR: Acid forms dimers via hydrogen bonds (1); Stronger attractions (1); Ethanoic acid 391 K vs ethanol 351 K (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Calcium and magnesium salts of the fatty acids (insoluble) (1). (ii) The long hydrocarbon chain (tail) (1). (iii) Micelles (1). (iv) Calcium and magnesium salts of detergents (sodium salts of sulphonic acids or ammonium salts) are soluble in water (1), so no insoluble scum forms and the cleansing action is not reduced (1).	1+1+1+2
20	X: ethene (1); $CH_2{=}CH_2 + H_2 \longrightarrow CH_3CH_3$ (1); Product: ethane (1); Ni is catalyst (1); Ethane has lower C% so burns cleanly (1).	1+1+1+1+1
	OR: Litmus: ethanoic acid red (1); Bicarbonate: ethanoic acid effervesces (1); Bromine water: ethene decolourised (1); Sodium: ethanol releases hydrogen (1); Ethane is left (1).	1+1+1+1+1

Mistakes to avoid:

- Counting covalent bonds wrongly: an alkane C_nH_{2n+2} has $3n + 1$ bonds (include all C-H and C-C bonds).
- Confusing ethanol and ethanoic acid: ethanol (C_2H_5OH) is an alcohol, ethanoic acid (CH_3COOH) releases CO_2 with carbonates; ethanol does not.

- Naming the chain from the wrong carbon count, e.g. calling $\mathrm{CH_3CH_2CHO}$ 'propanol' instead of propanal; the suffix comes from the functional group (-ol, -al, -one, -oic acid).
- Writing an addition reaction for a saturated hydrocarbon; saturated compounds undergo substitution, unsaturated ones undergo addition.
- Saying that detergents form micelles but soaps do not, or that soap works well in hard water; soaps give scum in hard water.
- Stating that diamond and graphite conduct electricity; only graphite does (it has free electrons).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND